

Project Harbour

Summary note on allocation risk in retirement

Monte Carlo simulation of retirement drawdown risk. 20,000 paths per scenario.

The short version

A retiree who spends 4% a year from a 60/40 portfolio runs out of money in 15% of paths when markets behave normally and nothing unexpected happens. Add clustered bad years, health bills and tax, and that rises to 25%. Add a bear market in the first year of retirement and it reaches 40%.

Allocation matters, but in a narrower way than most people expect. With static spending, ruin risk is lowest between 50% and 80% stocks and barely moves inside that range. Holding no stocks is the worst choice by a wide margin. Holding only stocks adds risk in the bad paths in return for a bigger pile in the good ones. The best mix also shifts with how stretched the plan is.

The bigger levers sit elsewhere. A guardrail spending rule halves the ruin rate. The first five years of returns decide more than the stock weight does. No allocation fixes a bad start on its own.

Measure	Result
Chance the money runs out (base case)	24.9%
Same, with a bear market in year one	40.0%
Same, with guardrail spending	11.9%
Chance of ending below the starting value	55.9%
Median ending wealth	\$773,259
Expected ending wealth	\$1,360,412

What was simulated

The portfolio starts at \$1,000,000 in today's money. It is split \$300,000 taxable, \$500,000 traditional and \$200,000 Roth. The retiree is 65 and wants \$40,000 a year after tax (a 4% rate) until age 95. The base mix is 60% stocks and 40% bonds. Every figure is in real terms, so inflation is already taken out.

Markets switch between three regimes. Expansion is the normal state and lasts close to ten years on average. A bear year is a sharp fall that is usually followed by a recovery, but can slide into stagflation. Stagflation is a long stretch of weak stock returns and falling bond prices, like the 1970s. Each regime has its own returns, volatility and stock and bond correlation. The long run mix sits close to US history. These numbers are my assumptions. They are not fitted estimates.

Regime	Share of years	Stocks mean	Stocks volatility	Bonds mean	Stock and bond correlation
Expansion	72%	11.0%	14.0%	2.5%	0.10
Bear	11%	-11.0%	24.0%	4.0%	-0.20
Stagflation	16%	3.0%	19.0%	-2.0%	0.60
Long run mix	All years	7.2%	17.8%	1.9%	0.14

Health shocks come in two kinds. A long term care episode can start from age 75, with a 1% yearly chance that rises to 5% by age 95. It costs \$60,000 a year for about two and a half years, and care costs rise 1.5% a year faster than inflation. About 45% of paths face an episode, at an average total of \$193,000. Smaller one off bills (3% a year, \$20,000 on average) can land at any age.

Three spending rules were tested. Static spending holds the real amount fixed. Guardrails cut spending by 10% when the withdrawal rate drifts 20% above its starting level, and give a 10% raise when it drifts 20% below. Spending stays between 70% and 150% of target. Variable percentage withdrawal (VPW) spends a share of what is left after the health bill, and that share rises as the years run down.

Tax is modelled across the three accounts. Traditional withdrawals count as income under a US single filer schedule. Taxable withdrawals pay capital gains tax on the gain only, plus a yearly tax on dividends. Roth withdrawals are free. Forced minimum withdrawals start at 73. Four withdrawal orders were compared.

The textbook answer is too kind

Chart 1 adds one problem at a time. With normal returns and no health bills, 15.2% of paths run out of money. Regime switching lifts that to 19.6%. The long run average return is identical in those first two steps. Only the shape of the risk changes, because bad years now arrive together and a retiree cannot wait them out. Health shocks take it to 24.9%. A bear market in year one takes it to 40.0%.

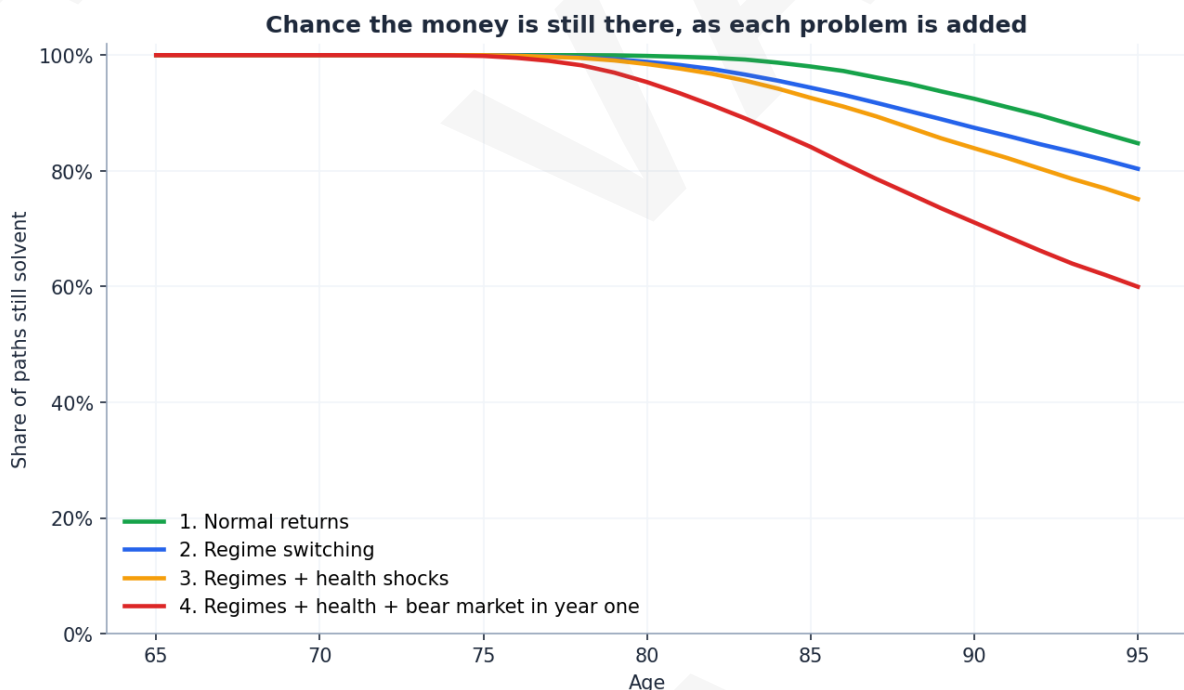


Chart 1. Share of paths still solvent at each age, as each problem is added.

Scenario	Chance of ruin	Chance of ending below start	Median ending wealth	Expected ending wealth
1. Normal returns	15.2%	54.0%	\$872,482	\$1,331,796
2. Regime switching	19.6%	51.3%	\$947,680	\$1,504,815
3. Regimes + health shocks	24.9%	55.9%	\$773,259	\$1,360,412

Scenario	Chance of ruin	Chance of ending below start	Median ending wealth	Expected ending wealth
4. Regimes + health + bear market in year one	40.0%	70.3%	\$273,597	\$875,891

A 25 point gap between the textbook case and the stressed case is not small. Anyone who tests a plan only against normal returns is measuring the wrong thing.

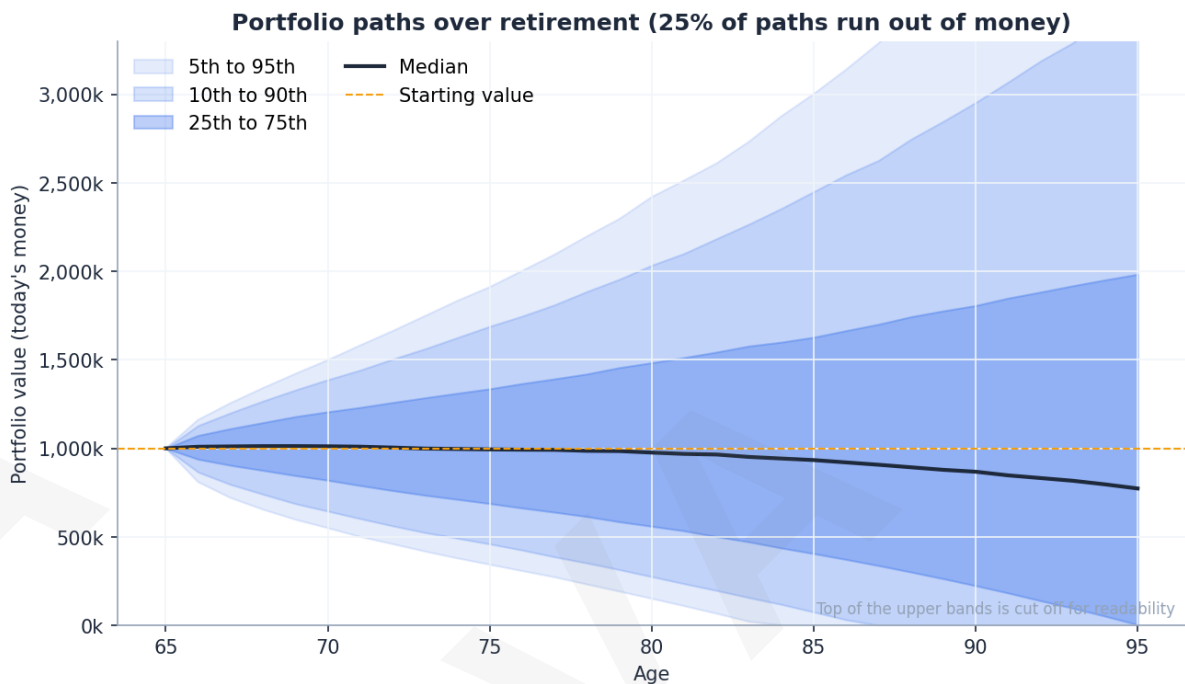


Chart 2. Portfolio value at each age in the base case. Bands show 50%, 80% and 90% of paths.

The fan in Chart 2 has a wide upper tail and a lower tail that reaches zero. The median path ends close to where it began (\$773,000), while the average is far higher (\$1.36 million) because a few paths do very well. Expected ending wealth is a poor guide for one retiree. The median and the ruin rate say more.

What it means for allocation risk

Chart 3 sweeps the stock share from 0% to 100% with static spending. Ruin risk falls steeply as stocks are added, from 62.4% with none to 30.8% at 30% and 24.9% at 60%. Then it flattens. From 50% to 80% stocks the chance of running out stays between 24.8% and 25.7%. Past 80% it rises again, to 27.0% at 100%.

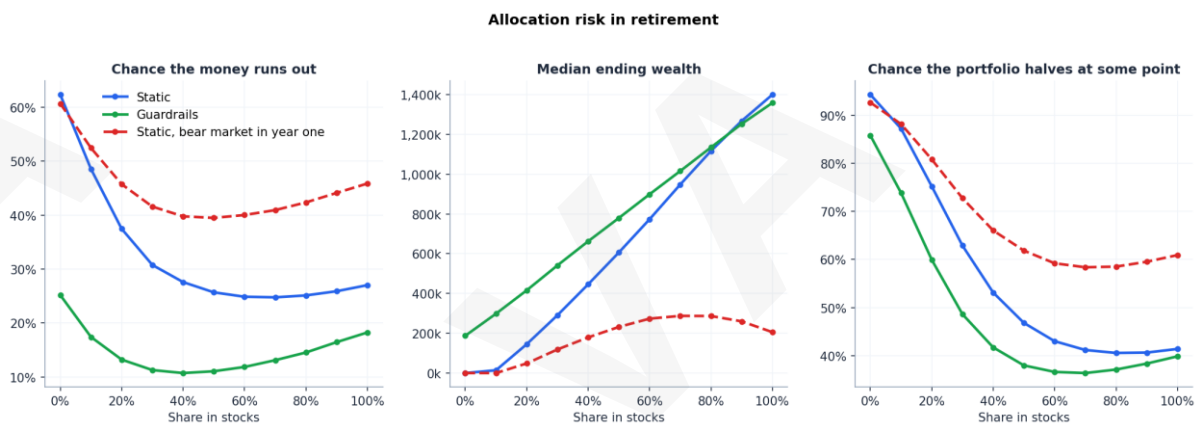


Chart 3. Ruin risk, median ending wealth and chance of the pot halving, by stock weight.

Mix	Static	Static with year one bear	Guardrails	Median ending wealth (static)
0% stocks	62.4%	60.7%	25.2%	\$0
20% stocks	37.5%	45.7%	13.2%	\$145,307
40% stocks	27.6%	39.8%	10.7%	\$446,109
50% stocks	25.7%	39.5%	11.1%	\$606,297
60% stocks	24.9%	40.0%	11.9%	\$773,259
70% stocks	24.8%	40.9%	13.1%	\$948,163
80% stocks	25.1%	42.4%	14.5%	\$1,116,835
100% stocks	27.0%	45.9%	18.3%	\$1,400,039

The all bond result is the one to remember. Bonds earn about 1.9% a year after inflation in this model, well short of a 4% withdrawal rate. Safety here means a slow and certain drain. Low volatility does not protect a plan that withdraws more than the portfolio earns.

The best mix moves when the start goes wrong. With a bear market in year one, ruin is lowest at 50% stocks (39.5%) and climbs faster on the equity side, to 45.9% at 100%. Median ending wealth tells the same story from the other direction. In the base case it is \$1.40 million at 100% stocks against \$0.77 million at 60%. In the bear start case the order flips, with \$206,000 at 100% against \$274,000 at 60%.

So the mix that minimises ruin is not the mix that maximises wealth. Which one is right depends on what the money is for. If spending must be met, the middle of the range is the safer answer. If leaving money behind matters more, a higher weight is a fair bet, as long as the bad paths are accepted too.

I also reran the sweep under four other sets of assumptions. In the three harder cases (lower returns, \$50,000 of spending, and a retirement at 60) the best static mix moved up to 80% or 100% stocks. A plan that draws more than a safe portfolio earns cannot afford the slow drain of bonds. It has to take equity risk to have a chance. A plan with room to spare can sit near 50% to 70% and lose very little. Guardrail spending pulled the best mix back to between 40% and 70% stocks, and to 50% or lower in four of the five cases. Guardrails beat static spending at every stock weight in every case.

Assumptions	Best static mix	Static ruin there	Best guardrail mix	Guardrail ruin there	All bond ruin (static)
Default	70%	24.8%	40%	10.7%	62.4%
Lower returns	80%	36.5%	50%	20.0%	79.5%
Normal returns model	70%	21.3%	40%	6.4%	65.8%
Spending \$50,000	100%	38.8%	70%	24.4%	93.6%
Retire at 60	80%	30.0%	50%	16.2%	80.0%

Drawdown risk is high at every mix. At 60% stocks the median path has a worst peak to trough fall of 52% (withdrawals included), and 43% of paths see the pot halve at some point. That is a feature of retirement and not a flaw in the model. Withdrawals make every fall deeper, and there are no new savings to repair it.

Sequence risk is bigger than allocation

Chart 4 splits the paths into fifths by the average real return in their first five years. The worst fifth averaged a loss of 3.8% a year, and two in three of those paths ran out of money. The best fifth averaged a gain of 11.9% a year, and only 4% ran out. The withdrawals were the same in both groups. What changed was the luck of the opening years.

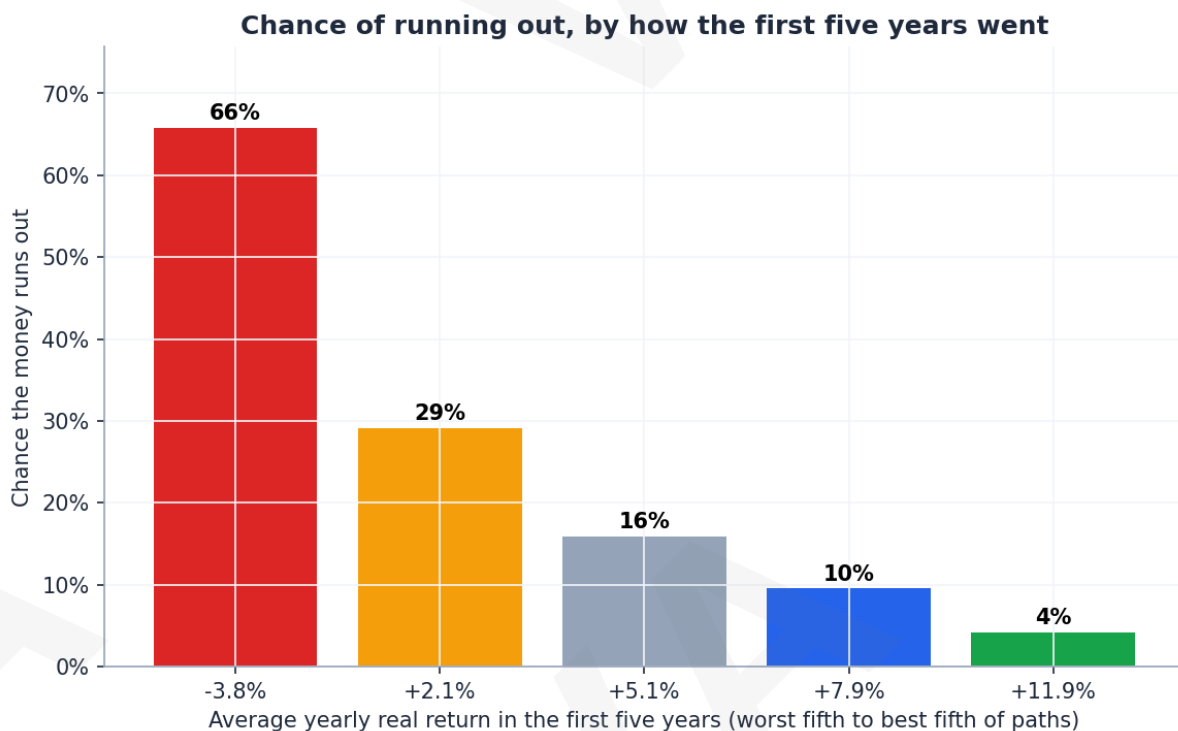


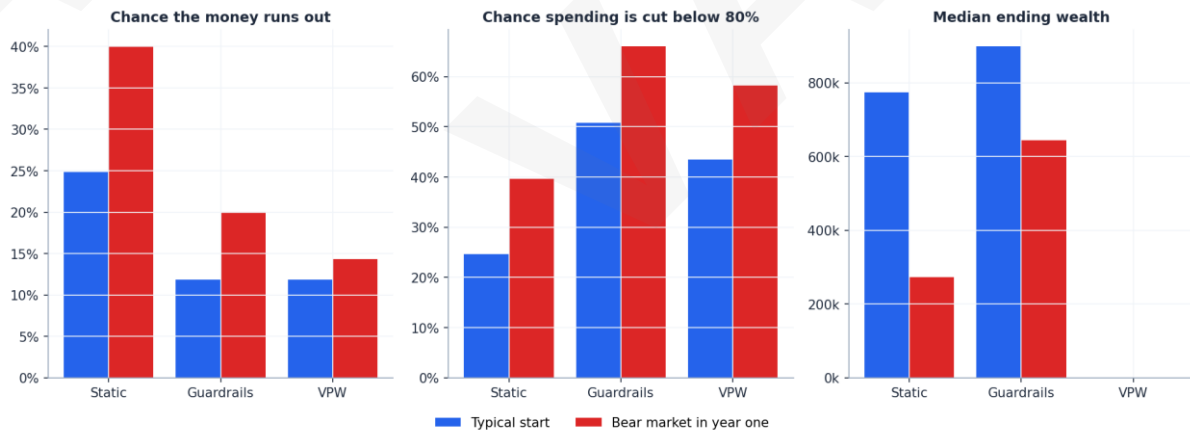
Chart 4. Chance of running out, by how the first five years of returns went.

This is why a bear market in year one moves the ruin rate by 15 points. It is also why the mix in the first five years deserves more thought than the mix at 85. A glide path that starts with less stock and adds more later was not tested here. It is the first thing I would test next.

Spending rules and tax

Chart 5 compares the three withdrawal rules. Guardrails cut the ruin rate from 24.9% to 11.9% at the same starting spending. VPW reaches 11.8%. Neither is free. Under guardrails, 51% of paths see spending cut below 80% of target at some point, against 25% for static spending (and static only falls short once the money has gone). VPW spends the pot down by design, so its ending wealth is close to zero. Its remaining ruin cases come from care bills after age 90, when little is left to pay them.

Dynamic withdrawal rules move the risk. They do not remove it.



VPW spends the pot down by design, so its ending wealth is close to zero. Judge it on spending, not on what is left.

Chart 5. Ruin risk, spending cuts and ending wealth for each withdrawal rule.

Rule	Chance of ruin	Chance spending below 80%	Median ending wealth	Median lifetime tax
Static	24.9%	24.6%	\$773,259	\$91,691
Static + year one bear	40.0%	39.7%	\$273,597	\$68,327
Guardrails	11.9%	50.8%	\$898,930	\$93,610
Guardrails + year one bear	20.0%	66.0%	\$643,746	\$76,560
VPW	11.8%	43.5%	\$0	\$76,584
VPW + year one bear	14.3%	58.2%	\$0	\$64,265

A rule that lets spending flex is worth more than any allocation change in this model. The price is a lower and less certain lifestyle in the bad years. Some retirees will accept that trade. Others will not, and the numbers should be read with that in mind.

Chart 6 compares withdrawal orders. Tax lifts the ruin rate from 20.6% to somewhere between 22.7% and 24.9%. The order matters by two points at most. Drawing from all three accounts in proportion did best (22.7%) and paid the least tax over a lifetime (about \$76,000). Filling the low brackets did worst (24.9%). At this wealth, the low brackets already cover most of the spending. Taking extra traditional money early gives up years of tax deferred growth, and the surplus that is reinvested pays tax on its dividends every year. A larger traditional balance would probably change that result, but I did not test it.

Withdrawal order changes ruin risk by a few points, not by a lot

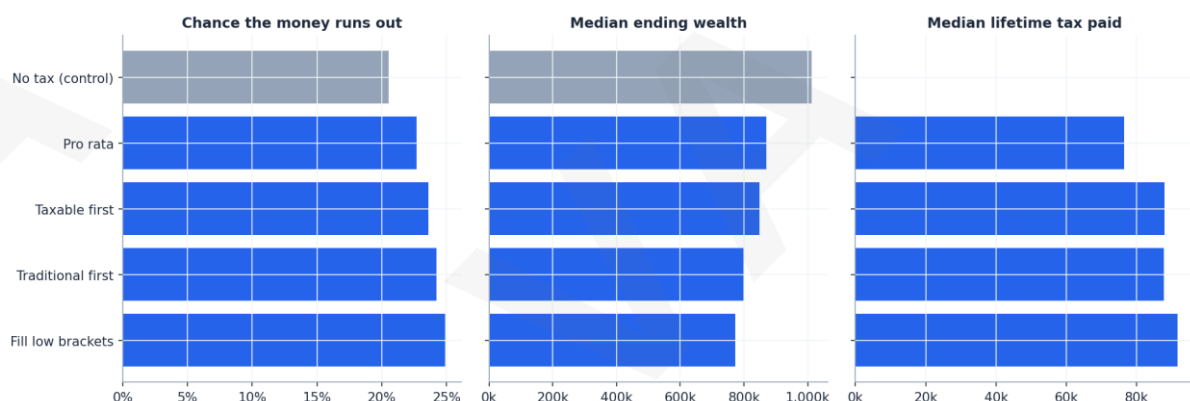


Chart 6. Ruin risk, ending wealth and lifetime tax for each withdrawal order.

Order	Chance of ruin	Median ending wealth	Median lifetime tax	Tax as share of withdrawals
No tax (control)	20.5%	\$1,011,451	\$0	0.0%
Pro rata	22.7%	\$870,360	\$76,459	7.0%
Taxable first	23.6%	\$849,520	\$87,850	7.7%
Traditional first	24.2%	\$799,645	\$87,815	7.6%
Fill low brackets	24.9%	\$773,259	\$91,691	7.0%

What could change these answers

Returns matter most. Lowering stock returns by 1.5 points and bond returns by 0.75 points lifts the base case ruin rate from 24.9% to 37.9%. Spending matters just as much. A \$50,000 budget instead of \$40,000 lifts it to 43.7%. Retiring at 60 instead of 65, with five more years to fund, lifts it to 31.1%.

The plan runs to a fixed age of 95 and nobody dies early. That overstates ruin for people who die sooner, and it also means every retiree who reaches 80 is exposed to a care bill. Care happens once at most. The same stock and bond mix sits in every account, so true asset location was not tested. Social Security, pensions and state tax are left out.

The regime numbers are judgement. A different set would move every figure here. The rankings held across the variants I reran. Guardrails beat static spending everywhere, all bond mixes failed everywhere, and the middle of the range beat the extremes for guardrails. The exact percentages should not be quoted as forecasts. Simulation noise is small. Three different random seeds moved the headline ruin rates by less than half a point.

What it implies

Allocation is a real choice with a wide, flat middle. Avoiding the extremes matters more than finding the exact point. All bond portfolios fail. A stretched plan needs more stocks, not fewer. A flexible spending rule is worth more than any weight change. The first five years are the dangerous window. And a plan that only passed a normal returns test has not been tested.

This is practice analysis. The figures are model output under stated assumptions. They are not forecasts and not personal advice.